

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

Finding Reasoning Paths :

- 1. **Epidermis ↔ Neural crest-derived structures ↔ Axial skeletal mesoderm-derived structures ↔ Radial artery . **
- 2. **Abduction of the forearm ↔ Radial nerve ↔ Radial artery . **
- 3. **Knock knock ↔ Penetrating knife wound ↔ Anatomical snuffbox ↔ Radial artery . **
- 4. **Thromboxane A2 (TXA2) ↔ Platelet activation ↔ Vascular hemorrhage ↔ Radial artery . **

Reasoning Process :

- 1. **Understanding the anatomical context : **
 - The **anatomical snuffbox** is an anatomic region located in the forearm , bordered by the styloid process of the radius , styloid process of the ulna , and the inferior head of the biceps brachii . It is part of the **axial skeletal mesoderm** .
 - The **radial artery** runs through the anatomical snuffbox , providing vascular supply to the forearm and hand .
- 2. **Analyzing the trauma mechanism : **
 - The teenager was involved in a penetrating knife wound in the **anatomical snuffbox region** . This area is anatomically close to the **radial artery** , which lies superficially and is palpable over this region .
 - Damage to the **radial artery** in this area could result in significant bleeding due to its rich blood supply and proximity to the wound site .
- 3. **Role of the vascular system : **
 - The **radial artery** is a branch of the **brachial artery** , which is a major artery in the upper limb . It gives off the **posterior interosseous artery** and the **palmar branch of the superficial radial cutaneous nerve** .
 - The **radical anastomosis** between the radial and ulnar arteries is a critical network for blood flow , but in this case , the injury is localized to the radial artery itself .
- 4. **Evaluating the nerve contributions : **
 - The **radial nerve** , which originates from the brachial plexus , is closely related to the radial artery . Damage to the radial nerve (e.g . , due to a blow to the wrist) could be a secondary issue in this case , but the primary source of bleeding is likely the radial artery .
- 5. **Pathophysiology of bleeding : **
 - The penetrating wound in the anatomical snuffbox would directly damage the **radial artery** , leading to severe bleeding . The injury to the **radical anastomosis** is also a consideration , as it can contribute to hemorrhage , but the primary source of the bleeding in this scenario is the radial artery itself .

Conclusion :

The **radial artery** , which is part of the **axial skeletal mesoderm-derived structures** , is the most likely cause of excessive bleeding . This conclusion is supported by the anatomical location of the injury (anatomical snuffbox) , the proximity of the radial artery to the wound , and the clinical presentation of severe bleeding .

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<answer>

Radial artery

</answer>

